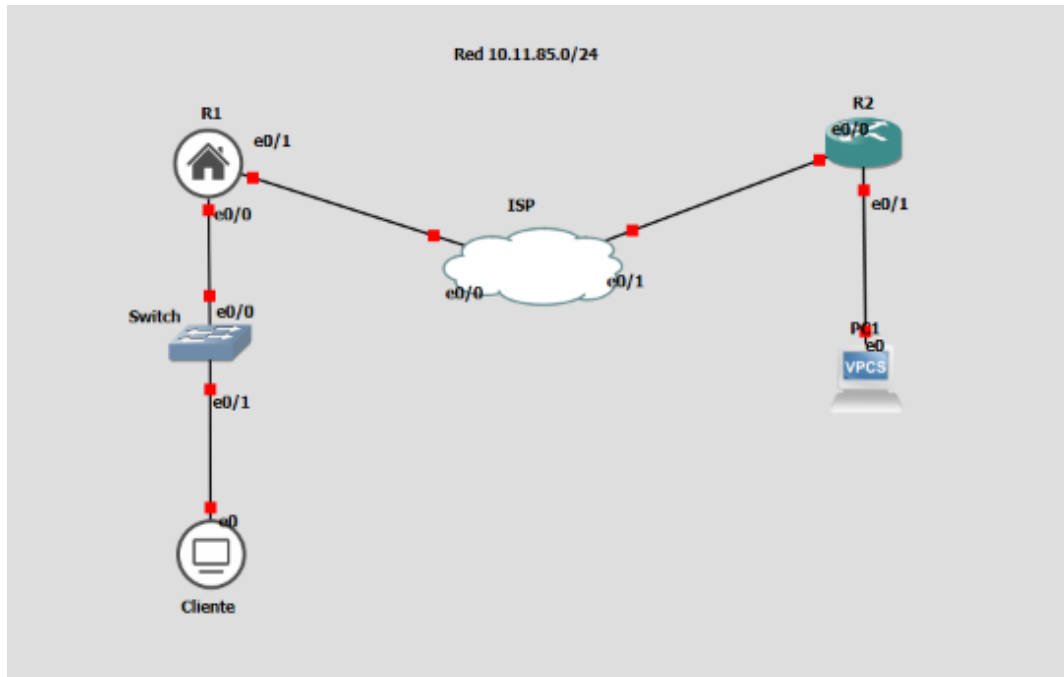


1. Objetivo

Implementar una VPN Client-to-Site punto a multipunto utilizando el protocolo L2TP (Layer 2 Tunneling Protocol) encapsulado sobre IPSec IKEv1. El cliente es una máquina Kali Linux que se conecta al servidor LNS (R2) a través de internet, atravesando un router NAT (R1). Una vez establecida la conexión, Kali recibe una IP virtual del pool del servidor y puede acceder a la LAN remota de R2 como si estuviera conectado localmente.

2. Topología de Red



Roles de cada dispositivo

Dispositivo	Rol	Descripción
Kali Linux	Cliente VPN	Inicia la conexión L2TP/IPSec desde la LAN de R1
R1	NAT Gateway	Da salida a internet al cliente Kali. NO termina el túnel VPN
ISP	Tránsito	Solo rutea. No participa en el túnel

R2 (LNS)	Servidor VPN	Termina el túnel L2TP/IPSec, autentica al cliente y asigna IP del pool
PC1	Host destino	Host en la LAN remota a la que el cliente VPN quiere acceder

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
R1	e0/0	10.11.85.1/26	LAN → Switch/Kali
R1	e0/1	10.11.85.66/30	WAN → ISP
ISP	e0/0	10.11.85.65/30	Hacia R1
ISP	e0/1	10.11.85.70/30	Hacia R2
R2	e0/0	10.11.85.69/30	WAN → ISP
R2	e0/1	10.11.85.129/26	LAN → PC1
PC1	e0	10.11.85.130/26	GW: 10.11.85.129
Kali	eth0	10.11.85.2/26	GW: 10.11.85.1 (R1)
Kali (VPN)	ppp0	10.11.85.202/32 (del pool)	IP virtual asignada por R2

3. Parámetros

IPSec IKEv1 (Fase 1 — ISAKMP)

Parámetro	Valor
Política	10
Cifrado	AES-256
Hash	SHA-256
Autenticación	Pre-Shared Key (wildcard 0.0.0.0 0.0.0.0)
Grupo DH	14 (2048-bit)
Lifetime	86400 segundos
Clave PSK	ITLA2024
NAT-T (NAT Traversal)	Habilitado — UDP 4500 (isakmp nat keepalive 20)

IPSec (Fase 2 — Transform Set)

Parámetro	Valor
Transform Set	TS-L2TP
Protocolo	ESP
Cifrado	AES-256
Integridad	HMAC-SHA-256
Modo	Transport (L2TP agrega su propio encapsulado)
Crypto Map	CM-L2TP → dynamic DYN-MAP

L2TP / PPP

Parámetro	Valor
Protocolo túnel	L2TP (vpdn-group L2TP-SERVER)
Template virtual	Virtual-Template 1
Pool de IPs	10.11.85.193 – 10.11.85.206 (/28)
Autenticación PPP	MS-CHAP v2
Cifrado PPP	MPPE auto
Usuario	kali

Contraseña

ITLA2024-VPN

4. Funcionamiento — L2TP/IPSec

L2TP/IPSec combina dos protocolos en capas. IPSec establece primero un canal cifrado seguro (Fase 1 y Fase 2), y dentro de ese canal IPSec, L2TP crea el túnel de capa 2 que permite negociar PPP para la asignación de direcciones IP al cliente.

Flujo de negociación

Fase	Protocolo	Descripción
1	IKEv1 Main Mode	Kali y R2 negocian parámetros ISAKMP y crean canal seguro (SA IKE). NAT-T activo por UDP 4500
2	IKEv1 Quick Mode	Negocian Transform Set IPSec en modo transport. SA IPSec establecida
3	L2TP	Dentro del canal IPSec, Kali abre túnel L2TP al LNS (R2) por UDP 1701
4	PPP/MS-CHAPv2	Autenticación del usuario (kali / ITLA2024-VPN)
5	IPCP	R2 asigna IP del pool a Kali. Interfaz ppp0 se levanta en Kali
6	Datos	Kali puede acceder a la LAN de R2 (10.11.85.128/26) via ppp0

Capas de encapsulación

```
Datos originales (IP: Kali → PC1)
↓ PPP encapsula
PPP frame
↓ L2TP encapsula
L2TP header + PPP frame  (UDP 1701)
↓ IPSec ESP (modo transport) cifra el UDP L2TP
ESP header + L2TP cifrado  (UDP 4500 por NAT-T)
↓ IP outer: Kali (NAT) → R2
IP outer: src 10.11.85.66 (R1 NAT) dst 10.11.85.67 (R2)
```

5. Scripts de Configuración

ISP

```
enable
configure terminal
hostname ISP

interface Ethernet0/0
description Hacia R1
ip address 10.11.85.65 255.255.255.252
no shutdown

interface Ethernet0/1
description Hacia R2
ip address 10.11.85.70 255.255.255.252
no shutdown

ip route 10.11.85.0 255.255.255.192 10.11.85.66
ip route 10.11.85.128 255.255.255.192 10.11.85.67
ip route 10.11.85.192 255.255.255.240 10.11.85.67

end
```

R1 — NAT Gateway

```
enable
configure terminal
hostname R1

interface Ethernet0/0
description LAN hacia Switch y Kali
ip address 10.11.85.1 255.255.255.192
ip nat inside
no shutdown

interface Ethernet0/1
description WAN hacia ISP
ip address 10.11.85.66 255.255.255.252
ip nat outside
no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.65

access-list 1 permit 10.11.85.0 0.0.0.63
ip nat inside source list 1 interface Ethernet0/1 overload

end
```

R2 — Servidor LNS

```
enable
configure terminal
hostname R2

interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.69 255.255.255.252
no shutdown

interface Ethernet0/1
description LAN hacia PC1
ip address 10.11.85.129 255.255.255.192
no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.70

aaa new-model
aaa authentication ppp VPN-AUTH local
aaa authorization network VPN-AUTH local

username kali password ITLA2024-VPN

ip local pool POOL-VPN 10.11.85.193 10.11.85.206

crypto isakmp policy 10
encryption aes 256
hash sha256
authentication pre-share
group 14
lifetime 86400

crypto isakmp key ITLA2024 address 0.0.0.0
crypto isakmp identity address

crypto ipsec transform-set TS-L2TP esp-aes 256 esp-sha256-hmac
mode transport

crypto dynamic-map DYN-MAP 10
set transform-set TS-L2TP
reverse-route
set pfs group2
exit

crypto map CMAP-L2TP 65535 ipsec-isakmp dynamic DYN-MAP
interface Ethernet0/0
crypto map CMAP-L2TP
exit

vpdn enable

vpdn-group L2TP-SERVER
accept-dialin
protocol l2tp
virtual-template 1
!
no l2tp tunnel authentication
```

```
interface Virtual-Template1
 ip unnumbered Ethernet0/1
 peer default ip address pool POOL-VPN
 ppp authentication ms-chap-v2 VPN-AUTH
 ppp authorization VPN-AUTH

end
```

VPCS — PC1

```
ip 10.11.85.130 255.255.255.192 10.11.85.129
save
```

Kali Linux — Cliente VPN (comandos principales)

```
echo "[*] Configurando eth1..."

sudo ip addr flush dev eth0
sudo ip addr add 10.11.85.2/26 dev eth0
sudo ip link set eth0 up

# Esto le dice al kernel: "Para llegar a cualquier cosa en 10.11.85.0/26, usa eth1"
sudo ip route add 10.11.85.0/26 dev eth0 src 10.11.85.2

sudo ip route add default via 10.11.85.1 dev eth0

echo "[+] eth0: 10.11.85.2/26 | GW: 10.11.85.1"

echo "[*] Ping a R1 (gateway)..."
ping -c 2 -W 2 10.11.85.1 && echo "[+] OK -> R1" || echo "FALLO -> R1, revisar
conexión física en GNS3"

echo "[*] Ping a R2 (servidor VPN)..."
ping -c 2 -W 2 10.11.85.67 && echo "[+] OK -> R2" || echo "FALLO -> R2, revisar NAT
en R1 y rutas en ISP"

echo ""
echo "[*] Instalando strongswan y xl2tpd..."
sudo apt-get update -y -qq
sudo apt-get install -y strongswan xl2tpd

echo "[*] Escribiendo /etc/ipsec.conf..."
sudo bash -c 'cat > /etc/ipsec.conf << EOF
config setup
    charondebug="ike 1, knl 1, cfg 0"
    uniqueids=no

conn L2TP-PSK
    keyexchange=ikev1
    authby=secret
    type=transport
    left=%defaultroute
    leftprotoport=17/1701
    right=10.11.85.69
    rightprotoport=17/1701
    rightid=10.11.85.69
    ike=aes256-sha256-modp2048!
    esp=aes256-sha256-modp2048!
    pfs=yes
    forceencaps=yes
    dpdaction=clear
    dpddelay=30s
```

```
dpdtimeout=120s
auto=add
EOF'
echo "[+] /etc/ipsec.conf OK"

echo "[*] Escribiendo /etc/ipsec.secrets..."
sudo bash -c 'cat > /etc/ipsec.secrets << EOF
%any %any : PSK "ITLA2024"
EOF'
sudo chmod 600 /etc/ipsec.secrets
echo "[+] /etc/ipsec.secrets OK"

echo "[*] Escribiendo /etc/xl2tpd/xl2tpd.conf..."
sudo bash -c 'cat > /etc/xl2tpd/xl2tpd.conf << EOF
[global]
port = 1701

[lac l2tp-vpn]
lns = 10.11.85.69
ppp debug = yes
pppoptfile = /etc/ppp/options.l2tpd.client
length bit = yes
EOF'
echo "[+] /etc/xl2tpd/xl2tpd.conf OK"

echo "[*] Escribiendo /etc/ppp/options.l2tpd.client..."
sudo bash -c 'cat > /etc/ppp/options.l2tpd.client << EOF
ipcp-accept-local
ipcp-accept-remote
refuse-eap
require-mschap-v2
noccip
mtu 1400
mru 1400
noipdefault
defaultroute
usepeerdns
connect-delay 5000
name kali
password ITLA2024-VPN
EOF'
echo "[+] /etc/ppp/options.l2tpd.client OK"

sudo mkdir -p /var/run/xl2tpd
sudo touch /var/run/xl2tpd/l2tp-control

echo ""
echo "[*] Reiniciando servicios..."
sudo systemctl stop xl2tpd 2>/dev/null || true
sudo systemctl stop strongswan-starter 2>/dev/null || true
sleep 2

sudo systemctl start strongswan-starter
sleep 3
sudo systemctl start xl2tpd
sleep 2
```

```
sudo systemctl is-active strongswan-starter && echo "[+] strongswan: ACTIVO" ||  
echo "strongswan: FALLO"  
sudo systemctl is-active xl2tpd && echo " xl2tpd: ACTIVO" || echo "[!]  
xl2tpd: FALLO"  
  
echo ""  
echo "[*] Levantando túnel IPsec hacia 10.11.85.69..."  
sudo ipsec up L2TP-PSK  
sleep 4  
  
echo "[*] Estado IPsec:"  
sudo ipsec status | grep -E "L2TP|ESTABLISHED|INSTALLED|ERROR" || echo " Revisar  
con: sudo ipsec statusall"  
  
echo ""  
echo "[*] Iniciando sesión L2TP (usuario: kali)..."  
sudo bash -c 'echo "c l2tp-vpn" > /var/run/xl2tpd/l2tp-control'  
  
echo "[*] Esperando que ppp0 se levante (15 segundos)..."  
for i in $(seq 1 15); do  
    sleep 1  
    if ip link show ppp0 &>/dev/null; then  
        echo "[+] ppp0 detectada en ${i}s"  
        break  
    fi  
    echo " ... ${i}s"  
done  
  
if ip link show ppp0 &>/dev/null; then  
    echo "[+] Agregando ruta hacia LAN de R2..."  
    sudo ip route add 10.11.85.128/26 dev ppp0 2>/dev/null && \  
    echo "[+] Ruta 10.11.85.128/26 via ppp0 OK" || \  
    echo "[*] Ruta ya existente"  
else  
    echo "ppp0 no se levantó. Ejecutar diagnóstico:"  
    echo " sudo journalctl -u strongswan-starter -n 40 --no-pager"  
    echo " sudo journalctl -u xl2tpd -n 40 --no-pager"  
    exit 1  
fi
```

6. Verificación

En R2 (servidor)

```
show vpdn session  
show vpdn tunnel  
show crypto isakmp sa
```

```
POOL-VPN          10.11.85.193    10.11.85.206      13      1      0
R2#sh vpdn session
```

```
L2TP Session Information Total tunnels 1 sessions 1
```

LocID	RemID	TunID	Username, Intf/ Vcid, Circuit	State	Last Chg	Uniq ID
32491	36853	39271	kali, Vi2.1	est	00:20:55	20

```
R2#sh vpdn tunnel
```

```
L2TP Tunnel Information Total tunnels 1 sessions 1
```

LocTunID	RemTunID	Remote Name	State	Remote Address	Sessn Count	L2TP Class/ VPDN Group
39271	52749	kali	est	10.11.85.2	1	L2TP-SERVER

```
R2#
```

```
R2#sh ip local pool
```

Pool	Begin	End	Free	In use	Blocked
POOL-VPN	10.11.85.193	10.11.85.206	13	1	0

```
R2#
```

```
R2#sh crypto isakmp sa
```

```
IPv4 Crypto ISAKMP SA
```

dst	src	state	conn-id	status
10.11.85.69	10.11.85.2	QM_IDLE	1017	ACTIVE

```
IPv6 Crypto ISAKMP SA
```

```
R2#sh crypto ipsec
```

```
show crypto ipsec sa
```

```
R2#sh crypto ipsec sa
```

```
interface: Ethernet0/0
```

```
  Crypto map tag: CMAP-L2TP, local addr 10.11.85.69
```

```
protected vrf: (none)
```

```
local  ident (addr/mask/prot/port): (10.11.85.69/255.255.255.255/17/1701)
```

```
remote ident (addr/mask/prot/port): (10.11.85.2/255.255.255.255/17/1701)
```

```
current_peer 10.11.85.2 port 4500
```

```
  PERMIT, flags={} 
```

```
#pkts encaps: 97, #pkts encrypt: 97, #pkts digest: 97
```

```
#pkts decaps: 97, #pkts decrypt: 97, #pkts verify: 97
```

```
#pkts compressed: 0, #pkts decompressed: 0
```

```
#pkts not compressed: 0, #pkts compr. failed: 0
```

```
#pkts not decompressed: 0, #pkts decompress failed: 0
```

```
#send errors 0, #recv errors 0
```

```
local crypto endpt.: 10.11.85.69, remote crypto endpt.: 10.11.85.2
```

```
plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb Ethernet0/0
```

```
current outbound spi: 0x0(0)
```

```
PFS (Y/N): Y, DH group: group14
```

```
inbound esp sas:
```

```
inbound ah sas:
```

```
inbound pcp sas:
```

```
outbound esp sas:
```

En Kali Linux (cliente)

```
ip addr show ppp0
(euni@kali)-[~]
$ ip addr show ppp0 | grep "inet "
```

inet 10.11.85.202 peer 10.11.85.129/32 scope global ppp0

```
ip route show
```

```
ping 10.11.85.129
```

```
(euni@kali)-[~]
$ ping -c 3 -W 2 10.11.85.129

PING 10.11.85.129 (10.11.85.129) 56(84) bytes of data.
64 bytes from 10.11.85.129: icmp_seq=1 ttl=253 time=3.66 ms
64 bytes from 10.11.85.129: icmp_seq=2 ttl=253 time=1.93 ms
64 bytes from 10.11.85.129: icmp_seq=3 ttl=253 time=1.95 ms

--- 10.11.85.129 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.930/2.513/3.657/0.808 ms
```

```
ping 10.11.85.130
```

```
(euni@kali)-[~]
$ ping -c 3 -W 2 10.11.85.130

PING 10.11.85.130 (10.11.85.130) 56(84) bytes of data.
64 bytes from 10.11.85.130: icmp_seq=1 ttl=61 time=4.26 ms
64 bytes from 10.11.85.130: icmp_seq=2 ttl=61 time=2.09 ms
64 bytes from 10.11.85.130: icmp_seq=3 ttl=61 time=1.67 ms

--- 10.11.85.130 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.670/2.672/4.261/1.136 ms
```

Verificación de establecimiento del Túnel

```
(euni@kali)-[~]
$ sudo journalctl -u xl2tpd -f
[sudo] contraseña para euni:
jun 29 00:45:53 kali pppd[60965]: rcvd [IPCP ConfNak id=0x2 <addr 10.11.85.20
1>]
jun 29 00:45:53 kali pppd[60965]: sent [IPCP ConfReq id=0x3 <addr 10.11.85.20
1>]
jun 29 00:45:53 kali pppd[60965]: rcvd [IPCP ConfAck id=0x3 <addr 10.11.85.20
1>]
jun 29 00:45:53 kali pppd[60965]: Script /etc/ppp/ip-pre-up started (pid 6097
2)
jun 29 00:45:53 kali pppd[60965]: Script /etc/ppp/ip-pre-up finished (pid 609
72), status = 0x0
jun 29 00:45:53 kali pppd[60965]: not replacing existing default route via 10
.11.85.1 with metric -1
jun 29 00:45:53 kali pppd[60965]: local IP address 10.11.85.201
jun 29 00:45:53 kali pppd[60965]: remote IP address 10.11.85.129
jun 29 00:45:53 kali pppd[60965]: Script /etc/ppp/ip-up started (pid 60975)
jun 29 00:45:53 kali pppd[60965]: Script /etc/ppp/ip-up finished (pid 60975),
status = 0x0
jun 29 00:55:14 kali xl2tpd[60886]: xl2tpd[60886]: Session 'l2tp-vpn' already
active!
jun 29 00:58:53 kali xl2tpd[60886]: xl2tpd[60886]: udp_xmit failed to 10.11.8
```

```
(euni@kali)-[~]
$ sudo ipsec up L2TP-PSK
sleep 4

generating QUICK_MODE request 1860029771 [ HASH SA No KE ID ID NAT-OA NAT-OA ]
sending packet: from 10.11.85.2[4500] to 10.11.85.69[4500] (476 bytes)
received packet: from 10.11.85.69[4500] to 10.11.85.2[4500] (492 bytes)
parsed QUICK_MODE response 1860029771 [ HASH SA No KE ID ID NAT-OA NAT-OA N((24576)) ]
selected proposal: ESP:AES_CBC_256/HMAC_SHA2_256/MDMP_2048/NO_EXT_SEQ
detected rekeying of CHILD_SA L2TP-PSK{12}
CHILD_SA L2TP-PSK{13} established with SPIs cc19b6e6_i 5a38e822_o and TS 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
connection 'L2TP-PSK' established successfully

(euni@kali)-[~]
$ sudo ipsec status | grep -E "L2TP|ESTABLISHED|INSTALLED|ERROR" || echo "Revisar con: sudo ipsec statusall"

L2TP-PSK{3}: ESTABLISHED 8 minutes ago, 10.11.85.2[10.11.85.2] ... 10.11.85.69[10.11.85.69]
L2TP-PSK{6}: REKEYED, TRANSPORT, reqid 1, expires in 51 minutes
L2TP-PSK{6}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
L2TP-PSK{7}: REKEYED, TRANSPORT, reqid 1, expires in 51 minutes
L2TP-PSK{7}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
L2TP-PSK{8}: REKEYED, TRANSPORT, reqid 1, expires in 51 minutes
L2TP-PSK{8}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
L2TP-PSK{9}: REKEYED, TRANSPORT, reqid 1, expires in 51 minutes
L2TP-PSK{9}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
L2TP-PSK{10}: REKEYED, TRANSPORT, reqid 1, expires in 52 minutes
L2TP-PSK{10}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
L2TP-PSK{11}: REKEYED, TRANSPORT, reqid 1, expires in 59 minutes
L2TP-PSK{11}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
L2TP-PSK{12}: INSTALLED, TRANSPORT, reqid 1, ESP in UDP SPIs: cab34835_i ea5108cb_o
L2TP-PSK{12}: 10.11.85.2/32[udp/l2f] == 10.11.85.69/32[udp/l2f]
```